

We have processors, then why a microcontroller?

Every other device that we use every day is controlled by a family of specialised processors called ‘microcontrollers’. They are processors designed to perform only specific tasks. The main aim behind the design of the micro-



Microcontrollers increasing interactivity

controller is to reduce the power requirement without compromising on functionality.

Use cases

Microcontrollers are used in places where tasks performed are predefined and rarely complex. With these devices, there always will be some input that

has some definite expected output after processing. They're used in video games, computer mouse, washing machine, cameras, microwave, cars, bikes, printers, landline telephone and mobile phones. Due to the high level of specificity of applications, these devices have RAM, ROM and peripherals integrated on the die of a single chip which helps reduce the size of the processor and cost.

Microprocessors are used where tasks performed need a wider range of capabilities such as scientific analysis, running servers, game and software development, photo editing etc. These tasks need a high number of resources such as RAM, ROM and processing power.

Let's compare some of the features of both these devices:

- **Speed of operation:** The clock speed of the microprocessor is 100-1000 times faster than the microcontroller. Generally, microcontrollers operate at a rate of 1 MHz to 50 MHz (excluding your smartphone) while most microprocessors operate above 1 GHz.
- **Processing abilities:** Microprocessors have much higher processing capabilities compared to microcontrollers as they perform complex tasks involving a combination of multiple microcontroller instructions.
- **Level of integration:** A microprocessor is a chip with only processing powers (contains only the processing unit commonly known as CPU - central processing unit) and it doesn't have other components like